

Students Performance Prediction

Saad Abdullah

1. Introduction

Predicting student performance during a course can play a crucial role in supporting educational interventions. By anticipating how students are likely to perform, instructors and institutions can provide timely assistance, enhancing learning outcomes. This study aims to explore machine learning techniques to predict final student grades based on their weekly in-course activities, such as quizzes, mini-projects, and peer reviews. The challenge lies not only in selecting the appropriate models but also in determining the most relevant features to improve predictive accuracy while preventing overfitting or bias.

This project primarily uses supervised learning techniques—specifically, K-Nearest Neighbors (**KNN**) and **Random Forest**—to make grade predictions. A significant aspect of the work involves assessing the impact of a highly correlated feature, ‘Week8_Total,’ which aggregates a student’s total marks from various weekly activities. Two model variations are created: one incorporating this feature and one excluding it, to evaluate its potential influence on overall predictions.

Key objectives include:

- Preprocessing and feature selection to optimize the dataset for machine learning models.
- Building and evaluating KNN and Random Forest models with cross-validation to ensure consistency.
- Investigating feature importance and understanding the influence of aggregated features on predictive accuracy.
- Comparing the two models based on multiple evaluation metrics, ensuring generalizability.

2. Data Processing

The dataset used for this study consists of **48 columns** and **107 rows**, capturing various student activities throughout a machine learning course. The final grade, representing the target variable, was influenced by several features such as quiz scores, mini-project performance, peer reviews, and student interaction with course content. Given that the dataset contained no missing values, there was no need for extensive preprocessing to handle incomplete data. Instead, the focus was on selecting relevant features and preparing them for the modelling process.

The initial step involved conducting a correlation analysis [Fig1] to measure the relationship between each feature and the final grade. Features with a correlation coefficient greater than 0.5 were prioritized, as they were more likely to contribute to model performance. This process helped in reducing unnecessary features, particularly those that had little impact on the prediction task. For instance, the feature ‘Week8_Total’, which combines multiple weekly activities into a single score, had a strong correlation with the final grade, but care was taken to avoid over-reliance on this single feature.

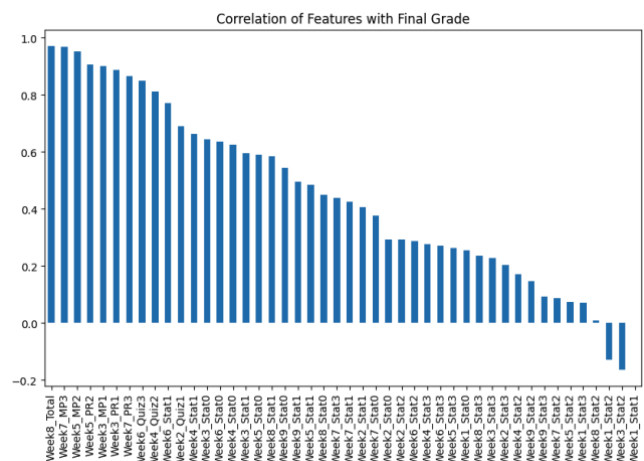


Figure 1: Correlation of Features with Final Grade

I selected features from three main categories: quizzes, mini-projects, and peer reviews. Each of these represented critical aspects of student performance across different weeks. Additionally, I included the 'Status 0' feature, which tracks student interaction with the course content, under the assumption that students who engaged more frequently with the material may likely perform better overall. This decision aligned with the general belief in educational settings that active participation tends to correlate with higher academic achievement.

In terms of feature selection techniques, I experimented with various methods to refine the dataset, although I did not include these in the final code. Recursive Feature Elimination (RFE) was also tested to identify and systematically eliminate less important features based on their contributions to model performance. Additionally, Permutation Importance was used with the Random Forest model, which involved shuffling individual feature values to assess their impact on model accuracy. While these techniques helped in understanding feature importance, they were not included in the final model.

Given the imbalanced nature [Figure 2] of the target classes (different grade categories), I used **stratified sampling** during the train-test split to ensure that all grade categories were adequately represented in both sets. This step helped the models generalize better across all grade categories.

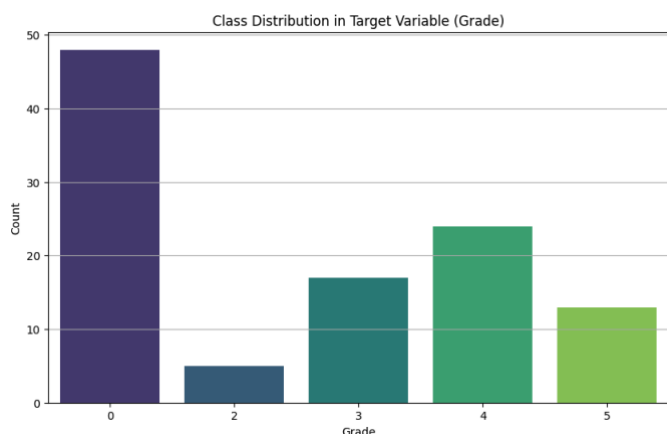


Figure 2: Class Distribution in Grades

Finally, I applied feature scaling to standardize the data, particularly because algorithms like KNN are sensitive to feature magnitudes. Standardization ensured that all features contributed equally to the model by bringing them to the same scale, with a mean of 0 and a standard deviation of 1. This was a critical step to avoid

skewing the results due to features with larger numerical ranges.

Two sets of models were trained: one with the highly correlated 'Week8_Total' feature and another without it. This approach allowed me to assess how much the inclusion of 'Week8_Total' influenced the model's performance compared to the more granular weekly features like individual quizzes, peer reviews, and mini-projects.

3. Proposed Approach

In this study, I utilized two machine learning algorithms: K-Nearest Neighbors (KNN) and Random Forest to predict student grades based on course activity.

K-Nearest Neighbors (KNN) was selected for its effectiveness in classification tasks and its non-parametric nature, which requires no training phase. This characteristic allowed for rapid implementation and adaptability to the dataset. I standardized the features to ensure uniform scaling, which is crucial for distance-based algorithms like KNN. Hyperparameter optimization was performed using Grid Search, exploring parameters such as the number of neighbors (n_neighbors) and weight function (weights). Model performance was evaluated using accuracy, precision, and recall metrics. To validate the model, I implemented K-fold cross-validation with a five-fold split to mitigate overfitting and combined this with a train-test holdout method for generalization. Additionally, I calculated feature importance using permutation importance to identify the most impactful features, reason for using permutation importance was because in KNN there is no concept of training.

For the Random Forest model, I chose this approach due to its robustness and ability to handle small datasets effectively. I trained it using two feature sets: one including the 'Week8_Total' feature and one excluding it. Similar to KNN, I standardized the features and utilized Grid Search with 5-fold cross-validation for hyperparameter tuning. Random Forest's inherent capability to assess feature importance facilitated the identification of key predictors. The evaluation metrics matched those of KNN for consistent performance comparison.

I initially considered using Decision Trees as well, but they exhibited significant overfitting, achieving 100% accuracy on the training set, which prompted me to exclude them from the final analysis. Given the numerous features, I applied Principal Component Analysis (PCA) for effective visualization of relationships among the features.

4. Data Analysis

Data visualization was instrumental in gaining insights into the relationships between features and student performance throughout the project. A variety of techniques were employed to analyze feature relevance and model behavior effectively.

Visualization and Evaluation:

- **Confusion Matrix:** Confusion matrices were utilized for both the KNN and Random Forest models to evaluate classification performance. They provided a clear picture of how well each model distinguished between different grade levels, highlighting instances where the models struggled, particularly with adjacent categories, such as differentiating between students scoring a 3 and those scoring a 5.
- **Principal Component Analysis (PCA):** PCA was employed to reduce the dataset's dimensionality, allowing for a visual representation of how well the models separated different grade categories. This analysis aided in understanding the underlying structure of the data. Refer to Figure [3,4,5, 6]

The models were assessed using several key metrics:

- **Accuracy:** Represents the percentage of correct predictions, providing an overall measure of model performance.
- **Precision:** Focuses on predicting higher grades by reducing false positives, ensuring students labeled as high performers truly have that capability.
- **Recall:** Recall helps us find students who need extra support, ensuring that those with lower grades are not missed by the model. A high recall means the model does a good job of identifying these students so they can get help when needed.
- **Cross-Validation:** To ensure the robustness of the models, K-fold cross-validation (with K=5) was implemented. This method enhanced the models' ability to generalize by validating performance

across multiple data subsets. Mean cross-validation accuracy was calculated for both models, providing a comprehensive overview of their effectiveness.

Feature Importance Analysis:

The Random Forest model, with its ensemble approach, offered valuable insights into feature importance. Key features, such as 'Week8_Total,' demonstrated a strong correlation with final grades and significantly impacted model predictions. To assess the influence of this feature further, I evaluated a version of the model without it, allowing for a direct comparison of performance and helping to identify the predictive power of other significant features.

5. Discussion of Results

The results showed that the inclusion or exclusion of specific features had a substantial effect on the predictive power of the models. The impact of the 'Week8_Total' feature was particularly noteworthy. When included, it boosted model performance significantly, particularly in predicting high-performing students. However, this came at the cost of possibly over-relying on this single aggregated feature, which could obscure other important variables, like individual weekly performances.

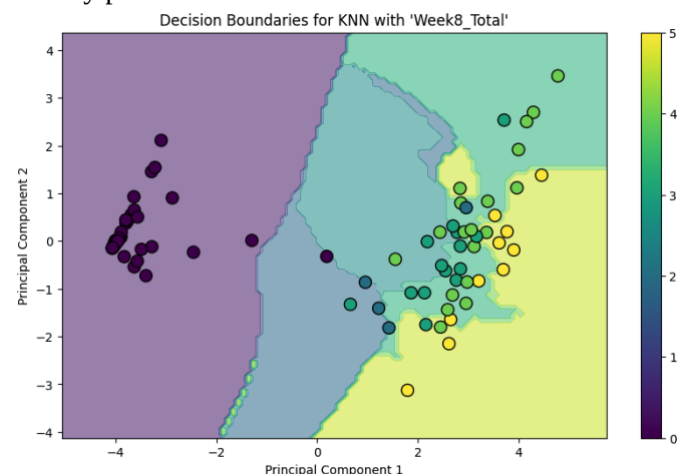


Figure 3: KNN with 'Week8_Total' Feature

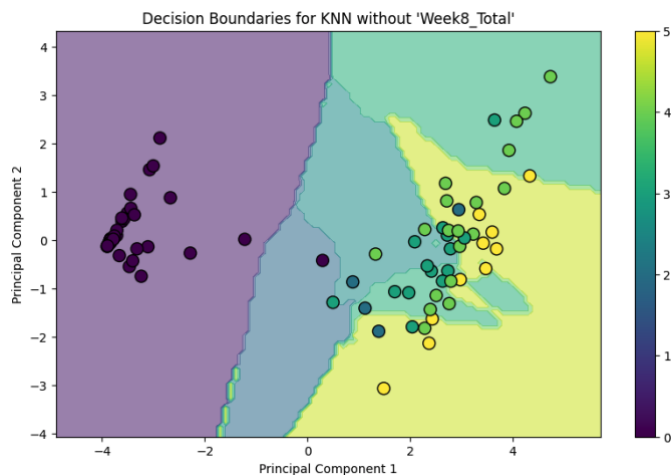


Figure 4: KNN without 'Week8_Total' Feature

Comparing KNN and Random Forest revealed key distinctions. Random Forest consistently outperformed KNN across most evaluation metrics, demonstrating a higher capacity for handling complex, non-linear relationships within the dataset. This was especially evident when the model was tasked with distinguishing between adjacent grade levels, where KNN struggled to maintain precision and recall.

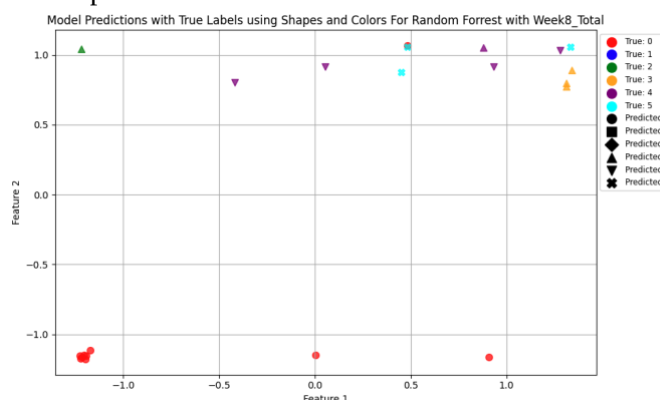


Figure 5: Random Forrest with 'Week8_Total' Feature

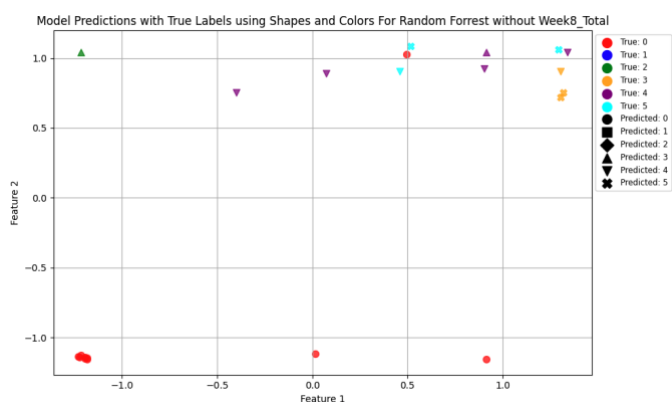


Figure 6: Random Forrest without 'Week8_Total' Feature

In models [Figure 3, 5] without 'Week8_Total', the performance gap widened, showing that Random Forest

was more adaptable to the loss of this feature. Its ability to draw on multiple decision trees to combine weaker signals from other features allowed it to maintain reasonable performance. KNN, on the other hand, was more dependent on this feature, and its accuracy suffered noticeably when it was removed.

In addition to feature importance, cross-validation further underscored the robustness of Random Forest. It consistently provided stable performance across different data folds, ensuring that the model was not overfitting. KNN, while still useful, displayed more variance and was less effective in the absence of highly correlated features.

6. Conclusion

The study demonstrated that supervised learning models, particularly Random Forest, can effectively predict student performance when given appropriate features. The inclusion of aggregated data like 'Week8_Total' improved accuracy but also presented risks of over-dependence. To mitigate this, the study evaluated model performance without this feature, revealing that while accuracy dropped, Random Forest remained resilient compared to KNN, which struggled without strong correlations.

The challenges encountered included the reliance on aggregated features, the limitations of KNN, and ensuring model generalization to unseen data. These were addressed by rigorously evaluating feature contributions and using robust validation techniques like stratified sampling and K-fold cross-validation.

The significance of feature selection and preprocessing was pivotal to the model's success. This study underscores the importance of developing models that can navigate complexity and adaptability in educational prediction tasks. Future exploration of more sophisticated methods, such as deep learning or hybrid approaches, may yield further performance enhancements as datasets become more intricate.

This study highlights that machine learning can be a valuable tool for enhancing student outcomes through timely and accurate performance predictions. For additional insights, including statistical evaluations and visual representations of model performance, please

refer to the detailed analysis present in the linked [Jupyter Notebook](#).

Reference:

Hashim, A. S., Awadh, W. A., & Hamoud, A. K. (2020). Student Performance Prediction Model based on Supervised Machine Learning Algorithms. *IOP Conference Series Materials Science and Engineering*, 928(3), 032019. <https://doi.org/10.1088/1757-899x/928/3/032019>

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